

G. W. Leibniz
SPECIMEN ANALYSEOS FIGURATAE

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Specimen Analyseos Figuratae in *Elementis* Geometriae

Analysin figuratam voco, quae modum praestat literis puncta
significantibus repraesentandi Figuras, et inveniendi atque
5 demonstrandi earum effectus et proprietates, ita ut non tantum
magnitudines, ut in Calculo Algebraico, sed situs ipsi
per novum hoc Calculi genus, directe exhibeantur. Specimen
autem hujus artificii edemus in *Elementis* Euclidis, et inter
procedendum assumemus Lemmata, Axiomata, Definitiones
10 aliasque propositiones, quibus indigebimus.

Quicquid autem in Decem prioribus libris Euclidis exponitur,
hoc intelligendum est *in uno esse plano*. Ne id perpetuo
admoneri opus sit.

Ad Lib. 1 Elem.

15 *Prop. 1.* Super data recta linea terminata *AB*. triangulum
aequilaterum *ABC* construere.

Specimen Analyseos Figuratae in *Elementis* Geometriae

I call figurate analysis that which affords a method of representing
figures by letters signifying points, and of discovering and demonstrating
their effects and properties, so that not only magnitudes, as
5R in the algebraic calculus, but also situations themselves are directly
exhibited by this new type of calculus. Here we will give a sample of
this technique¹ on the *Elements* of Euclid, and in the process we will
take up the lemmas, axioms, definitions, and other propositions we
need.

10R Now whatever is explained in the first 10 books of Euclid, this is to
be understood *to be in one plane* — so we won't need to be constantly
reminded of this.²

Regarding Book I of the Elements

15R **PROPOSITION 1.** Over a given straight line with endpoints *AB*, to
construct the equilateral triangle *ABC*.

¹Latin *artificii*. For Leibniz, 'art' or 'artifice' contrasts with labour and ingenuity. Cf. DAS par. 3 about Euclid: 'a work more of toil than of method and art, although now and then an art to the process seems to be concealed'.

²Leibniz believed that space is essentially 3-dimensional.

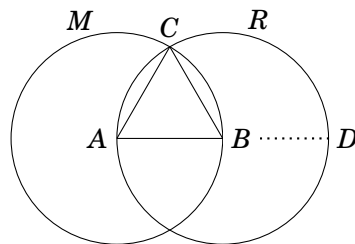


Figure 1

(1) $AC = AB$ ex hypothesi. (2) $BC = BA$ ex hyp. Et quod his
satisfacit est C . (3) Sit $AM = AB$. (4) Fiat (per postulat. 1) $\overline{M}$
circumferentia circuli centro A intervallo AB (per definitionem
1). (5) Sit $BR = BA$. (6) Fiat $\overline{R}$ circumf. circuli centro B
intervallo BA (ut ad 4). (7) Jam quoddam R est M (per Lemma
1) (8) seu $\overline{R}$ et $\overline{M}$ sibi occurrunt (ex. 7. per defin. 2). (9) Datis
lineis $\overline{M}$ et $\overline{R}$ (per 4 et 6) sibi occurrentibus (per 7, 8) habetur
eorum occursus (per *postulatum* 2) nempe R quod est M . (10)
Id vero est C (per 1 et 3 ac per 2 et 5). (11) Habetur ergo C .
10 (12) Jam datur AB (ex hyp.). (13) Ergo habetur ABC . Qu. Er.
Fac.

(1) $AC = AB$ by hypothesis. (2) $BC = BA$ by hypothesis. And C is that
which satisfies these. (3) Let $AM = AB$. (4) Then (by Postulate 1³) $\overline{M}$
becomes the circumference of a circle with centre A and interval⁴ AB (by
Definition 1). (5) Let $BR = BA$. (6) Then $\overline{R}$ becomes the circumference
5R of a circle with centre B and radius BA (as in 4). (7) Now some R is
 M (by Lemma 1). (8) That is, $\overline{R}$ and $\overline{M}$ meet each other⁵ (from 7 by
Definition 2). (9) Given curves $\overline{R}$ and $\overline{M}$ (by 4 and 6) meeting each
other (by 7, 8), one has their meeting (by Postulate 2), namely an R
which is an M . (10) But this is C (by 1 and 3 and by 2 and 5). (11)
10R Therefore we have C . (12) AB is already given (by hypothesis). (13)
Therefore we have ABC . Q.E.F.⁶

³Leibniz collected two lists of postulates, definitions, and axioms in the margins of the first two manuscript pages. We have provided the two lists in two boxes below.

⁴Latin *intervallo*. This frequently indicates a circle's radius.

⁵Leibniz crossed out *se secant* and replaced it with *occurrunt*. In 9 he replaced *intersectio* with *occursus*.

⁶*Quod Erat Faciendum*, i.e. 'that which was to be made'. Here Leibniz had 'Qu. Er. Fac.'. We standardize Leibniz's various forms of this expression to 'Q.E.F.' throughout the essay. Similarly we standardize 'Quod erat demonstrandum' to 'Q.E.D.' throughout.

Def. 1. Circumf. centr. interv. lib. 1. pr. 1 art. 4.

Postul. 1. Dato centro et intervallo circulum describere.

Def. 2. Occursus art. 8.

Postul. 2. Datis occurrentibus habetur occursus art. 9.

Def. 3. Intra circulum esse, art. 16.

Postul. 3. Recta produci potest ex uno termino ad distantiam quantamvis art. 17. 26.

Axiom. 1. Totum majus parte art. 20.

Def. 4. Extra circ. esse art. 21.

Ax. 2. Continuum quod intra et extra figuram est ejus circumferentiae occurrit art. 24.

Def. 5. Recta est via brevissima inter extrema sive distantia duorum punctorum. Art. 32.

Ax. 3. Omnes partes nullam partem communem habentes aequantur Toti. Art. 34.

Ax. 4. Quod movetur non est simul in pluribus locis. Art. 38.

Definition 1: Circumference, centre, interval. Book 1, Proposition 1, item 4.

Postulate 1: To describe a circle with a given centre and interval.

Definition 2: The meeting. Item 8.

Postulate 2: Given things that meet, one has their meeting. Item 9.

Definition 3: Being inside a circle. Item 16.

Postulate 3: A line can be extended from one endpoint to any distance whatever. Items 17, 26.

Axiom 1: The whole is greater than the part. Item 20.

Definition 4: Being outside a circle. Item 21.

Axiom 2: A continuum that is inside and outside a figure meets its circumference. Item 24.

Definition 5: A straight line is the shortest path between extremes, i.e. the distance of two points. Item 32.

Axiom 3: All parts, having no common part, equal the whole. Item 34.

Axiom 4: What moves is not in multiple places simultaneously. Item 38.

Lemma 1. Si duo circuli MA ex A et RA ex B habeant mutuo centrum in alterius circumferentia B in $\overline{M}$ et A in $\overline{R}$ eorum circumferentiae $\overline{M}$ et $\overline{R}$ sibi occurrent alicubi in C .

LEMMA 1. If two circles, MB^7 from A and RA from B , each have their centre in the other's circumference, B in $\overline{M}$ and A in $\overline{R}$, their circumferences meet someplace, at C . [See Figure 1.]

Iisdem quae antea positis, (14) $BA = BA$ (per se). (15) Ergo quoddam R est A (per 5). (16) Itaque quodd. R est intra circulum $A\overline{M}$ (per defin. 3, nam *punctum intra circulum esse* dicimus cujus distantia a centro est minor radio). (17) Producatur AB ex B in D (per postulatum 3) (18) ut sit $BD = BA$ (per Lemma 2). (19) Ergo $AD = AB + BD$ (per Lemma 3). (20) Ergo $AD \supset AB$ (totum parte per Axiom. 1). (21) Ergo D est extra $A\overline{M}$ (per def. 4, nam *punctum extra circulum esse* dicimus cujus distantia a centro est major radio). (22) Jam D est R (per 18 et 5). (23) Ergo qu. R est extra $A\overline{M}$. (24) Itaque (per 16 et 23) qu. R est in $\overline{M}$ (per Ax. 2, omne continuum $\overline{R}$ quod intra et extra figuram $A\overline{M}$ est, esse et in ejus circumferentia $\overline{M}$. *Schol.* Unde etsi vi formae ex puris

Supposing the same things as before, (14) $BA = BA$ (by itself). (15) Therefore some R is A (by 5). (16) And so some R is inside the circle $A\overline{M}$ (by Definition 3, since we say *a point is inside a circle* if its distance to the centre is less than the radius). (17) Let AB be extended from B to D (by Postulate 3) (18) so that $BD = BA$ (by Lemma 2). (19) Therefore $AD = AB + BD$ (by Lemma 3). (20) Therefore $AD \supset AB$ (the whole [is greater] than the part, by Axiom 1). (21) Therefore D is outside $A\overline{M}$ (by Definition 4, for we say *a point is outside a circle* if its distance from the centre is greater than the radius). (22) But D is R (by 18 and 5). (23) Therefore, some R is outside $A\overline{M}$. (24) And so (by 16 and 23) some R is on $\overline{M}$ (by Axiom 2, every continuum $\overline{R}$ which is both inside and outside the figure $A\overline{M}$, is also on its circumference $\overline{M}$. *Scholium:* From here, although nothing would follow formally from

⁷ MA in the manuscript, seemingly by mistake, though it is also possible that he intended ' MA from A and RB from B '.

particularibus nil sequatur, tamen vi materiae in continuis ex 16 et 23 sequitur 24.)

(25) Ergo (ex 24 per def. 2. ad 8) $\overline{M}$ et $\overline{R}$ sibi occurrunt. Q. E. D.

5 *Lemma 2.* Recta AB ex centro B (imo puncto quovis intra Circulum) produci potest ut et circumferentiae circuli $\overline{R}$ alicubi occurrat in D .

(26) Produci enim potest ad distantiam quantamvis (per postul. 3. ad 17). (27) Ergo ad E sic ut sit $BE \sqcap BA$. (28) Ergo
10 (per def. 4. ad 21) recta producta est extra circulum $B\overline{R}$. (29) Eadem est in circulo ad ejus centrum B (per def. 3. ad 16).
(30) Ergo (per Ax. 2. ad 23) occurrit ejus circumferentiae alicubi in D .

15 Coroll. Lemmatis 2. Recta AD transiens per punctum B intra circulum, circulo bis occurrit in A et D . Nam AB producta ex B (recedendo ab A) circumferentiae occurrit alicubi in D ,
per Lemm. 2. Et DB producta ex B (recedendo a D) circulo occurret alicubi in A .

20 *Lemma 3.* Si tria puncta $A.B.D.$ sint in recta distantia duorum quorundam ex ipsis, AB coincidit ipsi $AB + BD$ summae distantiarum tertiae D ab ipsis A, B .

pure particulars, materially, in continua, 24 follows from 16 and 23.⁸⁾

(25) Therefore, (from 24 by Definition 2 at 8) $\overline{M}$ and $\overline{R}$ meet each other. Q.E.D.

5R **LEMMA 2.** Line AB from centre B (indeed from any point inside the circle) can be extended so that it also meets the circumference of the circle $\overline{R}$ someplace, at D .

(26) For it can be extended to an arbitrarily large distance (by Postulate 3 at 17). (27) Therefore [it can be extended] to E such that $BE \sqcap BA$. (28) Therefore (by Definition 4 at 21), the line has been extended outside the circle $B\overline{R}$. (29) The same line is in the circle at its centre B (by Definition 3 at 16). (30) Therefore (by Axiom 2 at 24⁹), it meets its circumference someplace, at D .

15R *Corollary of Lemma 2.*¹⁰ A line AD passing through a point B inside a circle meets the circle twice, at A and D . For the line AB extended from B (receding from A) meets the circumference someplace, at D , by Lemma 2. And DB extended from B (receding from D) will meet the circle someplace, at A .

20R **LEMMA 3.** If three points A, B, D ¹¹ are on a line, the distance of some two of them, say AD , coincides with $AB + BD$, the sum of the distances of a third one B from A, D .¹²

⁸A *scholium* is a sort of interpretive marginal note or explanatory comment added to a primary text. They were commonplace in the scholastic period. This one is a reference to the form-matter distinction applied to syllogisms; the form is the argument structure or type; the matter is the actual contents or subject. Also in connection with syllogisms, 'particular' premises ('some A is B ') were said not to have formal consequences. So Leibniz is noting that 24 is not inferred from the logical structure of 16 and 23 alone, but results after his Axiom 2 about the continuum is additionally assumed.

⁹The manuscript has '23'.

¹⁰In the manuscript, this is written in the margin beside Lemma 3.

¹¹The manuscript has ' $A.B.D$ ', which appears to be an intrusion of a notation with dots that Leibniz uses elsewhere but has otherwise avoided in this essay.

¹²Leibniz wrote two versions of this lemma and proof. In the first version, A and B are the endpoints, while in the second version, A and D are the endpoints. He cancelled the first version's proof and the second version's statement. In addition, he mixed up B and D in two places in the first version's statement. We have opted to make A and D the endpoints for both statement and proof. Finally, at 'of a third one B from A, D ', Leibniz actually wrote 'tertia D ab ipsis A, B ', apparently assigning 'tertia' to an elided feminine noun

(31) Tria puncta sunt in recta (ex Hyp.). (32) *Recta AD* est via brevissima seu distantia inter extrema *A* et *D* (per def. 5). (33) Ergo punctum *B* in recta minus distat ab extremo *A*, quam extrema *A*, *D* inter se. (34) Et $AB + BD = AD$ (omnes partes nullam partem communem habentes toti, per ax. 3). (35) Est autem AB distantia inter *A* et *B* et BD inter *B* et *D* (per def. 5. ad 32). (36) Superest ut ostendamus tribus punctis in recta existentibus capi posse rectam quae duobus ex illis terminetur tertium vero includat. (37) Ponamus enim punctum aliquod rectam cui insunt percurrere. (38) Id ipsa attinget successive (per axioma 4). (39) Esto ergo *A* primum, *B* secundum, *D* tertium. (40) Ergo portio rectae quam percurreret inter *A* et *D* terminabitur ipsis *A* et *D*, comprehendet autem *B*.

(31) The three points are on a line (by hypothesis). (32) The *line AD* is the shortest path, or the distance, between the extremes *A* and *D* (by Definition 5). (33) Therefore the point *B* on the line is less distant from the extreme *A* than the extremes *A* and *D* are from each other. (34) And $AB + BD = AD$ (all the parts having no common part [are equal] to the whole by Axiom 3). (35) But AB is the distance between *A* and *B*, and BD the distance between *B* and *D* (by Definition 5 at 32). (36) It remains for us to show that when three points exist on a line, one can get a line which ends in two of them while including the third. (37) Indeed let us suppose that some point runs along the line which they are in [*insunt*¹³]. (38) It will reach them successively (by Axiom 4). (39) Therefore let *A* be the first, *B* the second, *D* the third. (40) Therefore the portion of the line it traverses between *A* and *D* will terminate in *A* and *D* but include *B*.

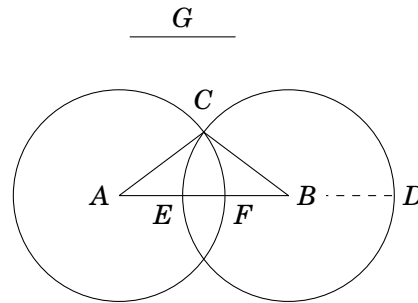


Figure 2

15 *Additamentum 1.* Si duo circuli $\overline{AM}$, $\overline{BR}$ aequalium radorum habeant radium, majorem dimidia distantia centrorum *A*. *B*. sibi occurrent in *C* extra rectam per centra.

15R *Addition 1.* If two circles, $\overline{AM}$, $\overline{BR}$, of equal radii have radius greater than half the distance of their centres *A.B.*,¹⁴ they will meet each other in *C* outside the line through their centres.

(perhaps 'distantia') instead of the neuter 'punctum' *D* as is really necessary to make sense of the clause. It is worth noting that he crossed out two different drafts of a marginal figure he drew showing three line segments with various permutations of *A*, *B*, *D* at the endpoints.

¹³This may or may not be used here in the technical sense that Leibniz developed for *inesse*, as part of his mereological calculus, around the period of this essay or not long after. Cf. SGL 'Some part of the doctrine about'.

¹⁴The '*A.B.*' was inserted later, which may explain the dot.

(41) Recta AB secat $\overline{R}$ bis in E et D (per corroll. Lemm. 2).
 (42) Et similiter $\overline{M}$ in F . (43) Sit E ex B versus A (44) et D ex B
 recedendo ab A (45) sitque F ex A versus B (46) AE erit minor
 AF (adde mox 58). (47) Nam quia (ob 43) E cadit inter B et
 5 A , vel B inter E et A (48) erit (per Lemm. 3) AE differentia
 inter AB et BE , (49) seu inter AB et AF ((50) quia $AF = BE$
 (ex Hyp.)). (51) Jam si $AF \sqcap AB$ (52) erit $AF = AB + AE$
 (per 48, 49). (53) Ergo $AF \sqcap AE$. (54) Sin $AB \sqcap AF$, erit
 $AB - AF = AE$ (per 48, 49). (55) Jam $AF \sqcap \frac{1}{2}AB$ (ex Hyp.).
 10 (56) Ergo $AE \sqcap \frac{1}{2}AB$. (57) Ergo $AF \sqcap AE$. (58) Utroque ergo
 modo erit $AE \sqcap AF$ ut asserebatur art. 46. (59) Rursus AD
 est major AF . (60) Nam $AD = AB + BD$ (per Lemm. 3) (61)
 $= AB + AF$ (quia $BD = AF$ ex hyp.). (62) Ergo quoddam R
 est intra $\overline{M}$ nempe E . (quia $AE \sqcap AF$ seu AM per 58). (63)
 15 Quoddam R est extra $\overline{M}$ nempe D (quia per 59 $AD \sqcap AM$ seu
 AF). (64) Ergo (per Ax. 2) Qu. R est in $\overline{M}$, ut C .

Additamentum 2. Super basi data AB . triangulum isosceles
 construere cujus crura AC vel BC sint magnitudinis datae G ,
 quam oportet esse majorem dimidia basi AB .

20 (65) Centris A et B (66) intervallis = G (per prop. 2. inde-
 pendentem ab hac demonstratam) (67) describantur circuli
 (postul. 1.) se secantes alicubi in C . Per additam. 1. erit
 $AC = G$ et $BC = G$. Quod erat fac.

25 *Prop. 2.* Ad datum punctum A ponere datae rectae BC ae-
 qualem rectam AG .

Solutio. (1) Jungatur AC . (2) Super qua Triang. Aequil. ACD

(41) The line AB intersects $\overline{R}$ twice, at E and D (by the corollary of
 Lemma 2). [See Figure 2.] (42) And similarly $\overline{M}$ at F . (43) Let E be
 [on the line] from B toward A (44) and let D be [on the line] from B
 moving away from A (45) and let F be [on the line] from A toward B ,
 5R (46) then AE will be less than AF (see 58 shortly). (47) Since (in view
 of 43) E falls between B and A , or B between E and A , (48) AE will
 be (by Lemma 3) the difference between AB and BE , (49) i.e. between
 AB and AF , ((50) since $AF = BE$ (by the hypothesis)). (51) Now if
 $AF \sqcap AB$, (52) we will have $AF = AB + AE$ (by 48, 49). (53) Therefore,
 $AF \sqcap AE$. (54) But if $AB \sqcap AF$, then $AB - AF = AE$ (by 48, 49). (55)
 10R Now $AF \sqcap \frac{1}{2}AB$ (by the hypothesis). (56) Therefore $AE \sqcap \frac{1}{2}AB$. (57)
 Therefore $AF \sqcap AE$. (58) Either way, therefore, $AE \sqcap AF$ as claimed in
 item 46. (59) In turn, AD is greater than AF . (60) For $AD = AB + BD$
 (by Lemma 3), (61) $= AB + AF$ (because $BD = AF$ ¹⁵ by hypothesis).
 15R (62) Therefore, some R is inside $\overline{M}$, namely E (since $AE \sqcap AF$, i.e. AM ,
 by 58). (63) Some R is outside $\overline{M}$, namely D (since by 59, $AD \sqcap AM$, i.e.
 AF ¹⁶). (64) Therefore (by Axiom 2), some R is on $\overline{M}$, say C .

20R *Addition 2.* Over a given base AB , to construct an isosceles triangle
 whose legs AC or BC are of a given magnitude G , which should be
 greater than half the base AB . [See Figure 2.]

(65) With centres A and B , (66) interval = G (by Proposition 2 demon-
 strated independently of this), (67) let circles be described (Postulate
 1) which intersect someplace, at C . By Addition 1 we will have $AC = G$
 and $BC = G$. Q.E.F.

25R **PROPOSITION 2.** At a given point A , to put a line AG equal to a given
 line BC . [See Figure 3.]

Solution. (1) Let AC be connected. (2) Let equilateral triangle ACD be

¹⁵The manuscript has ' $BD = BF$ ' but the argument calls for $BD = AF$.

¹⁶' AM ' in the manuscript.

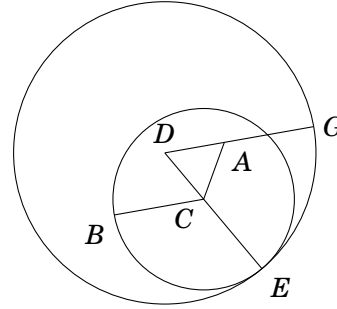


Figure 3

construatur (per 1. prim.). (3) Centro C . radio CB describatur
circumferentia circularis BE (per postul. 1.) (4) cui recta
 DC producta ex C (per postul. 3) (5) occurret alicubi in E
(per prop. 1. Lemm. 2). (6) Centro D radio DE describatur
5 Circulus (postul. 1) (7) cui recta DA producta ex A occurret
alicubi in G (per dict. Lemm. 2). (8) Erit $AG = BC$. (9) Nam
 $DC + CE = DE$ (per 5 et Lemm. 3 ad prop. 1.) (10) $= DG$
(per 6 et 7) (11) $= DA + AG$ (per 7 et Lemm. 3 ad prop. 1.)
(12) $= DC + AG$ (per 2). (13) Ergo (per 9 et 12) $AG = CE$ (14)
10 $= BC$ (per 3. 4).

Schol. ad prop. 2. Analysis qua constructio haec inveniri
potest, talis est. Ad punctum A ponenda est recta aequalis
rectae positae ad punctum C . Recta autem posita ad punctum
 C intelligi potest non tantum BC , sed et alia quaecunque ut
15 CE , ipsi CB aequalis, seu ex C ad circumferentiam circuli
 CBE ducta. Cum ergo puncta A et C debeant tractari eodem
modo, etiam recta AC tractanda est sic, ut respectu C tractetur
quemadmodum tractata est respectu A . Quaeratur ergo
punctum aliquod D eodem modo se habens ad A et C , quod fit
20 super AC construendo triangulum aequilaterum vel isosceles
 ADC . Ducta recta ex D per C producta occurret circulo in E ;

constructed upon it (by 1, I¹⁷). (3) Let a circular circumference BE be
described with centre C , radius BC (by Postulate 1), (4) which the line
 DC extended from C (by Postulate 3) (5) meets someplace, at E (by
Proposition 1, Lemma 2). (6) Let a circle be described with centre D ,
5 radius DE (Postulate 1), (7) which the line DA extended from A meets
someplace, at G (by the assertion in Lemma 2). (8) Then $AG = BC$. (9)
For $DC + CE = DE$ (by 5 and Lemma 3 of Proposition 1). (10) $= DG$
(by 6 and 7), (11) $= DA + AG$ (by 7 and Lemma 3 of Proposition 1). (12)
 $= DC + AG$ (by 2). (13) Therefore (by 9 and 12) $AG = CE$, (14) $= BC$
10R (by 3, 4).

Scholium to Proposition 2. The analysis by which this construction can
be discovered is like this: A line [segment] is to be put at point A equal
to the line put at point C . By a line put at point C one can understand
not only BC but also any other, such as CE , equal to CB , i.e. drawn
15 from C to the circumference of the circle CBE . Since, therefore, points
 A and C should be treated in the same way, the line AC should also be
treated in such a way that with respect to C it is treated just as it has
been treated with respect to A . Therefore, let some point D be sought
relating in the same way to A and C , which is made by by constructing
20R an equilateral or isosceles triangle ADC over AC . The line drawn from
 D , extended through C , will meet the circle in E ; in the line from D

¹⁷'1. prim' in the manuscript, literally '1 of the first', an elliptical reference to Proposition 1, Book I of Euclid's *Elements*. More such references follow.

ipsi DE sumatur aequalis DG in recta ex D producta per A ,
erit et $AG = CE$, quia eodem modo invenitur G respectu A ,
quo E respectu C .

Prop. 3. Duabus datis rectis A et BC majore ab ea detrahere
5 BE aequalem ipsi A .

extended through A let DG be taken equal to DE , and we will have
 $AG = CE$, because G is found in the same way with respect to A as E
with respect to C .

PROPOSITION 3. Given two lines, A and a larger one BC , from $[BC]$
5R cut off BE equal to A . [See Figure 4.]

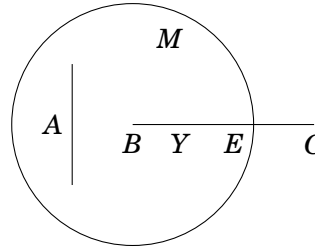


Figure 4

(1) Fac $BD = A$ (per 2. prim.) (2) et fac $\overline{M}$ ut sit $BM = BD$
(per postul. 1.). (3) Jam BC est intra $\overline{M}$ in B (ut ad art. 16.
prop. 1.) (4) extra $\overline{M}$ in C ((5) quia $BC \cap A$ ex hyp. (6) ergo
 $\cap BM$ per 1. 2.) (6) Ergo (per Ax. 2) BC occurrit ipsi $\overline{M}$ alicubi
10 in E . (7) Ergo BE pars BC . (8) Et $BE = BD = A$.

(1) Form $BD = A$ (by 2, I¹⁸) (2) and form $\overline{M}$ such that $BM = BD$ (by
Postulate 1). (3) Now BC is inside $\overline{M}$ at B (as in item 16 of Proposition 1)
(4) and outside $\overline{M}$ at C ((5) because $BC \cap A$ by hypothesis, (6) therefore
 $\cap BM$ by 1, 2). (6) Therefore (by Axiom 2) BC meets $\overline{M}$ someplace, at
10R E . (7) Therefore BE is a part of BC . (8) And $BE = BD = A$.

Def. 6. Si A sit aequale parti ipsius B , A minus dicetur B majus. Lib. 1. pr. 3.
art. 6.

Def. 7. Anguli rectilinei aequales dicuntur si congrui sunt. Prop. 4. Art. 3.

Ax. 5. Datis punctis ad quae anguli figurae consistunt figura data est. Art. 9.
Potest censi postulatum.

Ax. 6. Quae eodem modo ex datis congruis (determinate) dantur congrua sunt.
Art. 11.

Definition 6: If A is equal to a part of B , then A is said to be *less*, B *greater*. Book
1, Proposition 3. Item 6.

Definition 7: Rectilinear angles are said to be *equal* if they are congruent. Proposi-
tion 4. Item 3.

Axiom 5: Given the points at which the angles of a figure stand, the figure is given.
[Proposition 4.] Item 9. This can be considered a postulate.

Axiom 6: Those which are given (determinately) in the same way from congruent
givens are congruent. [Proposition 4,] item 11.

Item sic:

Also thus:

¹⁸Leibniz wrote '3. prim.', but it seems he meant Proposition 2.

(3) Sit $\bar{Y} \infty BC$. (4) Erit $BY + YC = BC$ (per Lemm. 3. ad prop. 1.). (5) $BC \sqcap A$ (ex hyp.). (6) Ergo qu. $BY = A$ (nam ex *definitione* 6. Majoris et Minoris, *Minus* est A cui qu. BY pars alterius, BC , quod *majus* dicitur, aequalis est). (7) Ergo qu. Y est M . (9) Quod sit E . Datur ergo E (per postul. 2.). (10) Ergo et $BE = A$ (per 6) (11) pars BC (per 4). Q. E. F.

Prop. 4. Si duo Triangula BAC , EDF , duo latera unius BA , AC duobus lateribus alterius ED , DF , respondens respondenti, aequalia habeant, et angulum unius A angulo alterius D aequalem, sub aequalibus rectis lineis contentum; tunc Triangulum triangulo congruum erit.

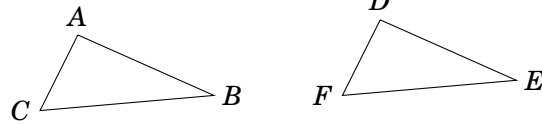


Figure 5

(1) Ponamus dari positione Angulos LAM et NDP , (2) aequales ex Hypoth. (3) ergo congruos. (Nam *anguli rectilinei aequales* definiuntur *def.* 7. qui congrui sunt.) (4) Et G magnitudinem AC et DF , (5) et H magnitudinem AB et DE . (6) Denique dari in quibus lateribus angulorum sumendae sint hae rectae, nempe AB in AM , AC in AL , DE in DP , DF in DN . (7) Dantur ergo puncta B , C , item E , F (per 3. primi). (8) Ergo A , B , C ; item D , E , F (per 1 et 7.). (9) Ergo Triangula ABC et DEF (nam datis punctis ad quae anguli figurae consistunt, figura data est per Ax. 5.). (10) Et quidem ambo eodem modo ex datis congruis determinate exhibentur (ex toto processu). (11) Ergo congrua sunt (per *Axioma* 6). Q. E. D.

(3) Let $\bar{Y} \infty BC$. (4) Then $BY + YC = BC$ (by Lemma 3 of Proposition 1). (5) $BC \sqcap A$ (by hypothesis). (6) Therefore some $BY = A$ (indeed by Definition 6. Greater and less. A is *less* when some part BY of the other one, BC , which is called *greater*, is equal to it). (7) Therefore some $BY = BM$ (by 6, 1, 2). (8) Therefore some Y is M . (9) Let it be E . Therefore E is given (by Postulate 2). (10) Therefore also $BE = A$ (by 6), (11) a part of BC (by 4). Q.E.F.

PROPOSITION 4. If two triangles BAC , EDF have two sides of the one, BA , AC equal to two corresponding sides of the other, ED , DF respectively, and angle A of the one equal to angle D of the other, subtended by the equal straight lines, then the one triangle is congruent to the other. [See Figure 5.]

(1) Let us suppose that the angles are given in position, LAM and NDP , (2) equal by the hypothesis, (3) hence congruent. (For *equal rectilinear angles* are defined in Definition 7 as those which are congruent.) (4) And [suppose] G [has] the magnitude of AC and DF (5) and H the magnitude of AB and DE . (6) Finally, it is given on which sides of the angles these lines are to be taken, namely AB in AM , AC in AL , DE in DP , DF in DN . (7) Therefore points B , C , likewise E , F are given (by 3, I). (8) Therefore A , B , C [are given]; likewise D , E , F (by 1 and 7). (9) Therefore, triangles ABC and DEF [are given] (for, given the points at which the angles of the figure stand, the figure is given by Axiom 5). (10) And indeed both are exhibited determinately in the same way from congruent givens (by the whole procedure). (11) Therefore they are congruent (by Axiom 6). Q.E.D.

Schol. Si quis superpositionem adhibeat res eodem redit, si enim data congrua fiant actu congruentia seu coincidentia per superpositionem, coincident etiam quae ex ipsis determinate dantur; alioqui non unum sed plura his datis satisfaciencia
 5 haberi possent, contra hypothesin. Unde Axiomatis sexti ratio intelligitur.

Porisma. Triangulum magnitudine et specie datur magnitudine datis angulo et lateribus eum comprehendentibus (per 1. usque ad 9). Nam ut positione detur tantum opus est
 10 angulum positione dari, et quae crura anguli, quibus magnitudinibus laterum assignentur, quae nihil in magnitudine et specie mutant, nam utrumque crus in angulo eodem modo se habet.

Schol. *Porisma* voco quod ex demonstratione colligitur, *corollarium* quod ex propositione.
 15

Prop. 5. Triangulum *ABC*, quod duo latera *AB*, *AC* aequalia habet, etiam angulos eorum *B*, *C* ad reliquum latus, *BC*, aequales habet.

Scholium. If one applies superposition, it comes to the same thing; for if the congruent givens become actually congruent [*actu congruentia*], i.e. coincident, through superposition, then those which are given determinately¹⁹ from them will also coincide; otherwise not one but several things satisfying these givens could be found, contrary to hypothesis. From this one understands the reason for the sixth axiom.

Porism. A triangle is given in magnitude and shape²⁰ if an angle and the sides enclosing it are given in magnitude (by 1 through 9). Indeed, for it to be given in position, one only needs an angle to be given in position, as well as which legs of the angle are assigned to which magnitudes of sides; this changes nothing in the magnitude and shape, since either leg relates in the same way at the angle.

Scholium. I call *porism*²¹ something inferred from a demonstration, and *corollary* something inferred from a proposition.

PROPOSITION 5. A triangle *ABC* which has two sides *AB* and *AC* equal also has their two angles *B* and *C* on the remaining side *BC* equal. [See Figure 6.]

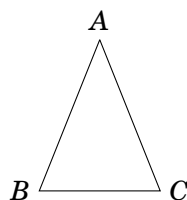


Figure 6

¹⁹Leibniz says that something given by a geometric procedure or construction is given *determinately* when it is unique among things satisfying the conditions, *semideterminately* when it is one of a finite (or discrete) set of things, and *indeterminately* when it is one of an infinite (or continuous) set of things satisfying the conditions. Cf. SGL ‘There is also the semidetermined’.

²⁰Leibniz’s word *species* can mean both ‘appearance’ in the sense of form or shape, and ‘species’ as a refinement of a ‘genus’.

²¹The word ‘porisma’ in Greek literally means *bonus* or *windfall*.

(1) Ponatur positione data BC . (2) Et data sit G , aequalis
 5 ipsis BA , CA (3) et *partes* ad quas cadere debet A (hoc est
 per *def.* 8. plano secto in duas partes a recta BC indefinite
 producta detur in qua parte debeat esse A). (4) Datur A si
 10 est possibilis, (5) nam centris B et C intervallo aequali G (per
 2. primi) (6) describuntur circuli, (postul. 1) (7) jam A est in
 circumferentia utriusque (per 2) si scilicet possibilis est. (8)
 Ergo circuli sibi occurrunt in A si A est possibilis. (9) Ergo
 (per postul. 2.) datur A . (10) Ergo (per ax. 5) datur ABC . (10)
 10 Ergo et anguli ABC , ACB (nam per ax. 7 dato aliquo dantur
 ejus requisita) (11) et quidem eodem modo (per processum
 expositum). (12) Ergo (per Ax. 6) congrui sunt anguli. (13)
 Ac proinde aequales. (Nam congrua sunt aequalia per ax. 8.)

Schol. Idem ostendi potuisset per superpositionem, si aliud
 15 sumtum fuisset triangulum huic congruum, DEF et nunc
 ABC applicaretur ipsi DEF ; nunc ACB ipsi DEF , ita nunc
 angulus ABC nunc angulus ACB congrueret eidem DEF , ergo
 congrui essent inter se.

Prop. 6. Triangulum ABC quod duos angulos B et C aequales
 20 habet, etiam latera AB , AC ad reliquum angulum A pertinen-
 tia aequalia habebit.

Demonstratur eodem modo, (1) quia dato latere BC , et angulis
 B , C et partibus ad quas debet esse A , datur A . (2) Nam data
 25 positione recta BC et angulo ad eam alterius rectae BA ,
 datur ipsa recta BA , indefinite (dato enim positione angulo
 latera dantur per ax. 7), eodem modo CA indefinite, (3) quae
 si possibilis est A se secabunt in A . (4) Datur ergo A , adeoque
 utraque eodem modo, ergo congruent. Q. E. D.

(1) Suppose BC is given in position. (2) Let G also be given, equal to
 BA , CA , (3) and the *side* on which [*partes ad quas*] A should fall (this
 is by Definition 8: the plane being cut into two parts by the line BC
 extended indefinitely, let it be given in which part A should be²²). (4)
 5R A is given if it is possible, (5) since, with centres B and C and interval
 equal to G (by 2, I), (6) circles are described (Postulate 1), (7) and now
 A is in the circumference of both (by 2), if it is possible, of course. (8)
 Therefore the circles meet each other at A if A is possible. (9) Therefore
 (by Postulate 2) A is given. (10) Therefore (by Axiom 5) ABC is given.
 10R (10) Therefore also the angles ABC , ACB (since by Axiom 7, when
 something is given, its requisites are given) (11) and indeed, in the
 same manner (by the procedure explained). (12) Therefore (by Axiom
 6) the angles are congruent. (13) And hence equal. (For congruents
 are equals by Axiom 8.)

15R *Scholium.* The same thing could have been shown by superposition:
 if some triangle DEF had been taken congruent to this one, and now
 ABC would be placed onto DEF , now ACB onto DEF ; thus now angle
 ABC , now angle ACB would be congruent to the same DEF ; therefore
 they would be congruent to each other.

20R **PROPOSITION 6.** A triangle ABC that has two angles B and C equal
 will also have the two sides AB and AC belonging to the remaining
 angle A equal.

This is demonstrated in the same way, (1) since, given side BC and
 angles B , C and the side on which A should be, A is given. (2) For
 25R line BC and the angle to it of another line BA being given in position,
 the line BA itself is given indefinitely (for the angle being given in
 position, by Axiom 7 the sides are given), in the same way CA [is given]
 indefinitely; (3) these will intersect each other in A if A is possible. (4)
 Therefore A is given, and thus both [sides are given] in the same way,
 30R therefore they are congruent. Q.E.D.

²²This definition did not make it into the marginal lists.

Schol. Idem potuisset demonstrari ex praecedenti. Ut et per superpositionem ad instar praecedentis.

Prop. 7. Si triangula ABC , DEF latera lateribus alterius aequalia habeant, respondens respondenti, Triangula erunt congrua.

5

Scholium. The same thing could have been demonstrated from the preceding. As also by superposition in the preceding fashion.

PROPOSITION 7. If triangles ABC , DEF have sides equal to the corresponding sides of the other, the triangles will be congruent. [See Figure 7.]

5R

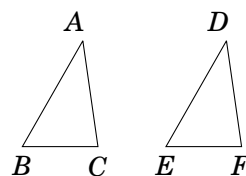


Figure 7

Demonstratur eadem Methodo, quia data positione uno latere unius trianguli, et uno alterius aequali, et reliquis datis magnitudine, circulos ex datorum positione laterum extremis, et magnitudinibus tanquam intervallis describendo dabitur utrumque triangulum, eodem modo, ergo per Ax. 6. congrua erunt.

10

This is demonstrated by the same method, since, with one side of one triangle being given in position, and [one side] of the other being equal to it, and the remaining ones being given in magnitude, then by describing circles from the extremes of the sides given in position and with the magnitudes as intervals, each of the triangles will be given, and in the same way; therefore by Axiom 6 they are congruent.

10R

Prop. 8. Datum Angulum BAC bisecare.

PROPOSITION 8. To bisect a given angle BAC . [See Figure 8.²³]

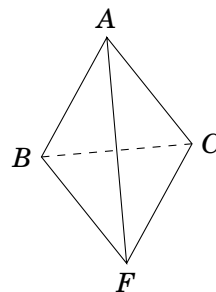


Figure 8

²³Leibniz originally had dashed lines AF , BF , and CF , but then he drew solid lines over them.

5R

10R

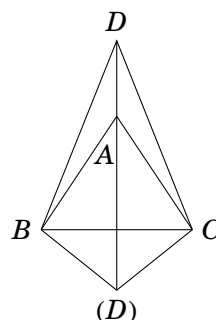


Figure 9

15R

The same will obtain if we construct any other isosceles triangle BFC at all over the base BC of the isosceles triangle ABC , by Addition 2 to Proposition 1. For the locus of all points relating in the same way to the two sides of the same angle or to the two extremes of the same line is a line passing through the two apices of the two isosceles triangles relating in the same way to the proposed angle or line.

²⁴'CB' in the manuscript.

Prop. 9. Datam rectam BC bisecare.

Quaerenda sunt duo puncta eodem modo se habentia ad B et C . Quod fiet si duo triangula isoscelia BAC , BDC qualiacunque construantur super basi BC , recta per angulos basi
5 oppositos seu per eorum apices ducta, basin bisecabit.

Schol. Euclides pro prop. 8. et 9. utitur triangulo aequilatero, sed praestat adhibere constructionem magis generalem.

PROPOSITION 9.²⁵ To bisect a given line BC . [See Figure 9.]

Two points should be sought relating in the same way to B and C . This will happen if two isosceles triangles BAC , BDC of any sort are constructed over the base BC ; the line drawn through the angles
5R opposite the base, i.e. through their apices, will bisect the base.

Scholium. Euclid uses an equilateral triangle for Propositions 8 and 9, but it is better to apply a more general construction.

²⁵The manuscript has 'Prop. 8' which repeats the number 8. It is not clear why, although Proposition 7 in Book 1 of Euclid is missing from Leibniz's presentation, so Leibniz's numbering does not align with Euclid at Propositions 7, 8, 9 above. It is possible that Leibniz considered his Proposition 8 to incorporate both of Euclid's Propositions 7 and 8.